Sahaj Bhadja

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Summary

Forward Deployed Engineer with an MS in CS and 2+ years applying **deep learning** in **Python**, from **simulation**-evaluated PyTorch models on 100K+ trajectory logs to production AI systems, translating technical results for clients and stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Shipped a multi-tenant **AI platform** on Azure SQL and Blob Storage for 500+ users, working with platform stakeholders and cutting dashboard latency from 1.2s to 350ms through query and index tuning.
- Built fault-tolerant **data pipelines** and LLM job orchestration in Python with FastAPI workers, validation, and failure isolation, processing 1K+ AI jobs at 3x throughput and 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Deployed **vision models** (GPT-4o Vision, Claude Vision, Qwen-VL) with OCR to read and grade handwritten work in production, serving 150+ instructors and 10K+ daily requests at 90%+ consistency.
- Worked with instructors to turn grading workflows into automated **AI adoption** through rubric generation, concurrent 1K+ file batches, and model fallback, cutting feedback turnaround 60% in Python.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with **geometric** HD-map encoding of road boundaries, lane directions, and connectivity aligned to a global frame, improving long-horizon planning 18% in simulation.
- Designed a multi-head attention encoder with distance-aware interaction weighting and evaluated it against three baselines in closed-loop **simulation**, improving closed-loop planning stability 15% while working within tight single-GPU compute limits.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Worked with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment, automating cross-platform data-sharing workflows that cut manual data entry 35% across Windows, Linux, and macOS.
- Led a redesign of AWS Glue and S3 **ETL** that normalized raw JSON, CSV, and Parquet data for analysis, boosting pipeline efficiency 68% and cutting storage costs 15%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered 21 statistical **features** such as IP presence, URL length, and suspicious keywords over a 650K+ sample dataset, building an end-to-end Python classification pipeline for phishing and malware detection.
- Trained and compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and **CNN** models, reaching 96%+ detection accuracy and presenting actionable insights on phishing and malware trends.

Skills

Languages: Python, C++, Java, SQL, TypeScript
Deep Learning: PyTorch, TensorFlow, Transformers, CNN, LSTM, Multi-Head Attention, Vision Models, Offline RL
ML & Data: Feature Engineering, Model Evaluation, Benchmarking, Random Forest, Hydra, Weights & Biases, ETL Pipelines
Simulation & Geometry: Closed-Loop Simulation, Spatial-Temporal Encoding, Coordinate Transforms, Trajectory Data
Cloud & Tools: AWS, Azure, GCP, Docker, Kubernetes, Git, Linux, FastAPI, REST APIs, PostgreSQL, CI/CD, OCR
Customer Delivery: Proofs of Concept, Client Presentations, Requirements Gathering, Technical Writing

Projects

Mesh-Based CFD Surrogate Model | *Python, PyTorch, PyTorch Geometric, OpenFOAM, Pandas*

- Designed a graph neural network **surrogate** that predicts surface pressure and drag on 3D vehicle meshes from public CFD datasets, validated against **OpenFOAM** simulations to rank design variants in seconds.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2*

- Fine-tuned and benchmarked BLIP, GIT, and ViT-GPT-2 **deep learning** models against a custom model on 10K images and 100K captions, raising caption accuracy from 81% to 96% behind real-time inference APIs.

A Detailed Exploration to Quantum Resistant Blockchain Technology Paper | *Simulation, Benchmarking*

- Co-authored an IEEE paper benchmarking a post-quantum protocol in **simulation** across 100–1,000 transaction workloads, sustaining 957.8 TPS at 224ms latency versus an ECDSA baseline of 968.3 TPS at peak load.